MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 11** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** comprises 6 questions of 1 mark each (5 MCQs and 1 Assertion-Reason).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based integrated unit of assessment of 4 marks.
7. Use of calculators is strictly prohibited. Draw neat, labeled geometric diagrams wherever required.

SECTION A: OBJECTIVE QUESTIONS (6 Marks)

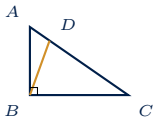
1. In ΔABC, D and E are on sides AB and AC such that $DE \parallel BC$. If $AD = 4$ cm, $DB = 4.5$ cm, and $AE = 8$ cm, then AC is:

(a) 17 cm (b) 9 cm (c) 16 cm (d) 18 cm

2. If $\Delta ABC \sim \Delta DEF$ such that $BC = 3$ cm, $EF = 4$ cm, and perimeter of $\Delta DEF = 24$ cm, perimeter of ΔABC is:

(a) 18 cm (b) 16 cm (c) 20 cm (d) 12 cm

3. In ΔABC, $AB = 5$ cm, $BC = 12$ cm, and $AC = 13$ cm. If $BD \perp AC$, then the length of BD is:



(a) $\frac{60}{13}$ cm (b) $\frac{65}{12}$ cm (c) 5 cm (d) 6 cm

4. If in ΔABC and ΔPQR, $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$, then:

(a) $\Delta PQR \sim \Delta CAB$ (b) $\Delta PQR \sim \Delta ABC$ (c) $\Delta CBA \sim \Delta PQR$ (d) $\Delta BCA \sim \Delta PQR$

5. In ΔABC, $DE \parallel BC$. If $AD = 3$ cm, $DB = 5$ cm, and $AE = 6$ cm, then EC is:

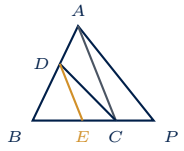
(a) 10 cm (b) 8 cm (c) 9 cm (d) 12 cm

6. **Assertion (A):** If $\Delta ABC \sim \Delta DEF$, the ratio of their corresponding angle bisectors equals the ratio of their corresponding sides.
Reason (R): Corresponding angles of similar triangles are equal, and angle bisectors create similar sub-triangles by AA criterion.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (4 Marks)

7. **[2 Marks]** In the given figure, $DE \parallel AC$ and $DC \parallel AP$. Prove mathematically that:



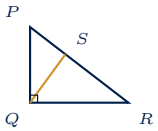
$$\frac{BE}{EC} = \frac{BC}{CP}$$

8. **[2 Marks]** In ΔABC, D is a point on side BC such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \cdot CD$.

SECTION C: SHORT ANSWER QUESTIONS (6 Marks)

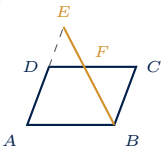
9. **[3 Marks]** In ΔABC, D, E, F are midpoints of sides AB, BC, CA respectively. Prove that ΔABC is divided into four congruent triangles, each of which is similar to ΔABC.

10. **[3 Marks]** In the given figure, PQR is a right-angled triangle at Q. S is a point on PR such that $QS \perp PR$. If $PQ = 15$ cm and $QR = 20$ cm, find the length of QS.



SECTION D: LONG ANSWER / HOTS QUESTIONS (10 Marks)

11. **[5 Marks]** In a parallelogram $ABCD$, E is a point on side AD produced and BE intersects CD at F. Prove that $\Delta ABE \sim \Delta CFB$. If $AB = 12$ cm, $CF = 8$ cm, and $FB = 6$ cm, find the length of line segment BE.

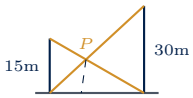


12. **[5 Marks]** In ΔABC, $AB = AC$. A circle through B touches side AC at its midpoint D and intersects AB at P. Prove that: (i) $AD^2 = AP \times AB$, (ii) $AB = 4AP$.

SECTION E: COMPETENCY / CASE STUDY QUESTION (4 Marks)

13. **Case Study 11: High-Voltage Transmission Cable Crossing Mechanics**

Two power transmission towers T_1 and T_2 of heights 15 m and 30 m stand vertically on horizontal ground separated by a span of 40 m. Tension guide wires run from the top of each tower to the anchored base of the opposite tower. Safety engineers must ensure the crossing clearance point P above ground meets high-voltage clearance norms.



(i) **[1 Mark]** State the algebraic formula for crossing height H in terms of tower heights h_1 and h_2.

(ii) **[1 Mark]** Does the horizontal distance between the towers affect clearance height H? Explain.

(iii) **[2 Marks]** Calculate the exact vertical clearance of crossing point P above the level ground.

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★